

Computer Architecture I



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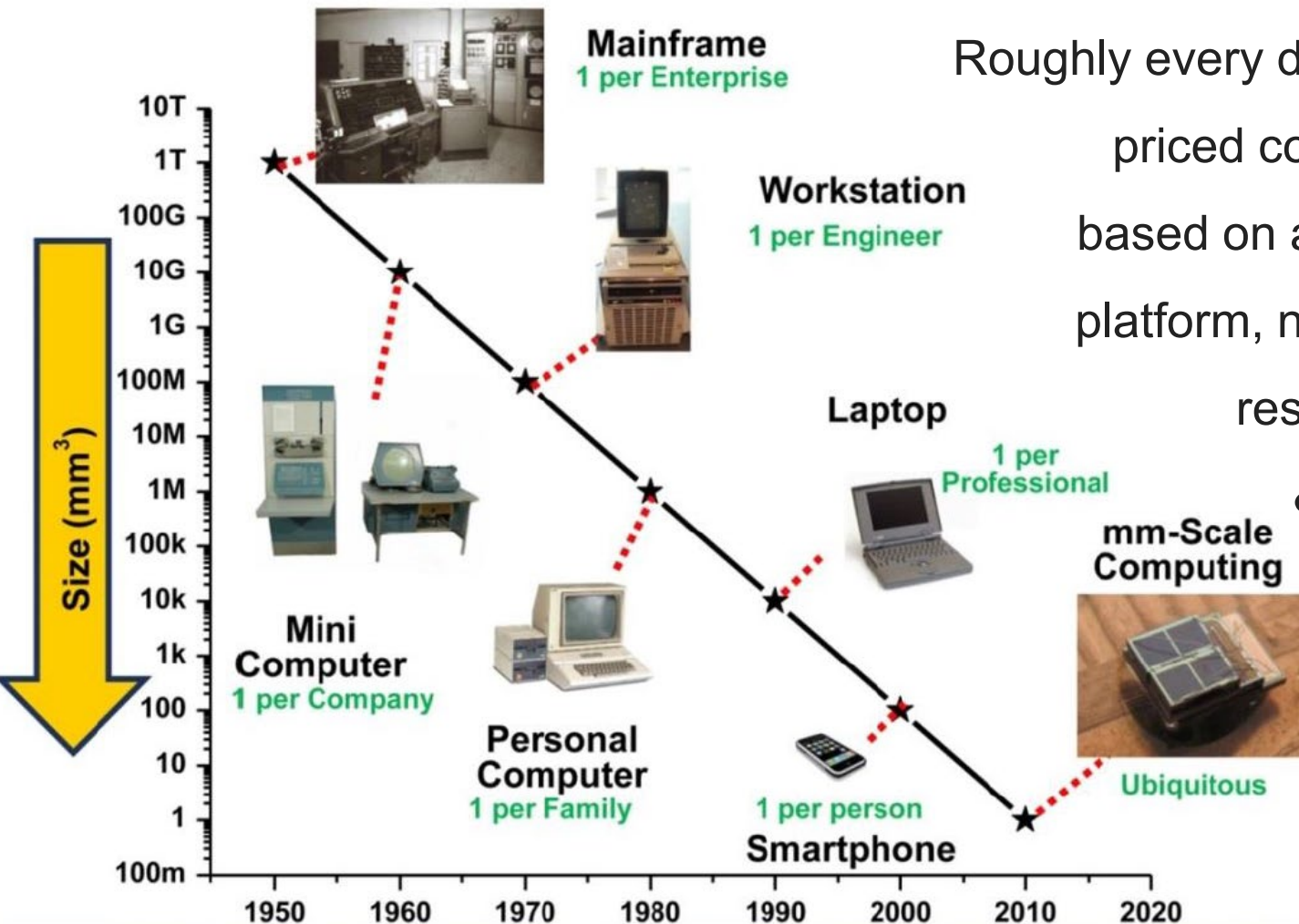


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for the birth and death

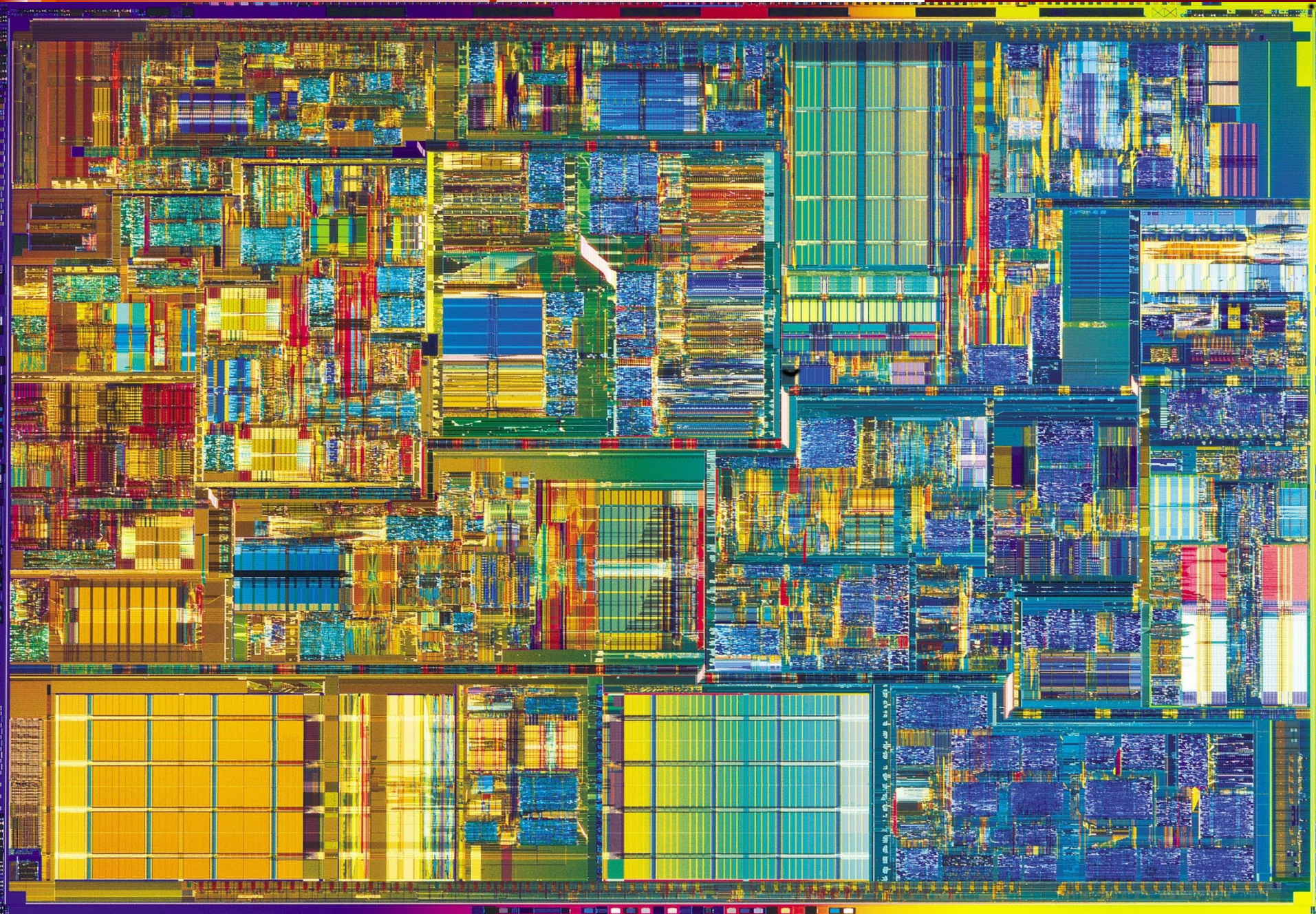
Bell's law of computer classes

Roughly every decade a new, lower priced computer class forms based on a new programming platform, network, & interface, resulting in new usage & the establishment of a new industry.



After "Rebooting the IT Revolution," 2015 (figure courtesy of D. Sylvester)

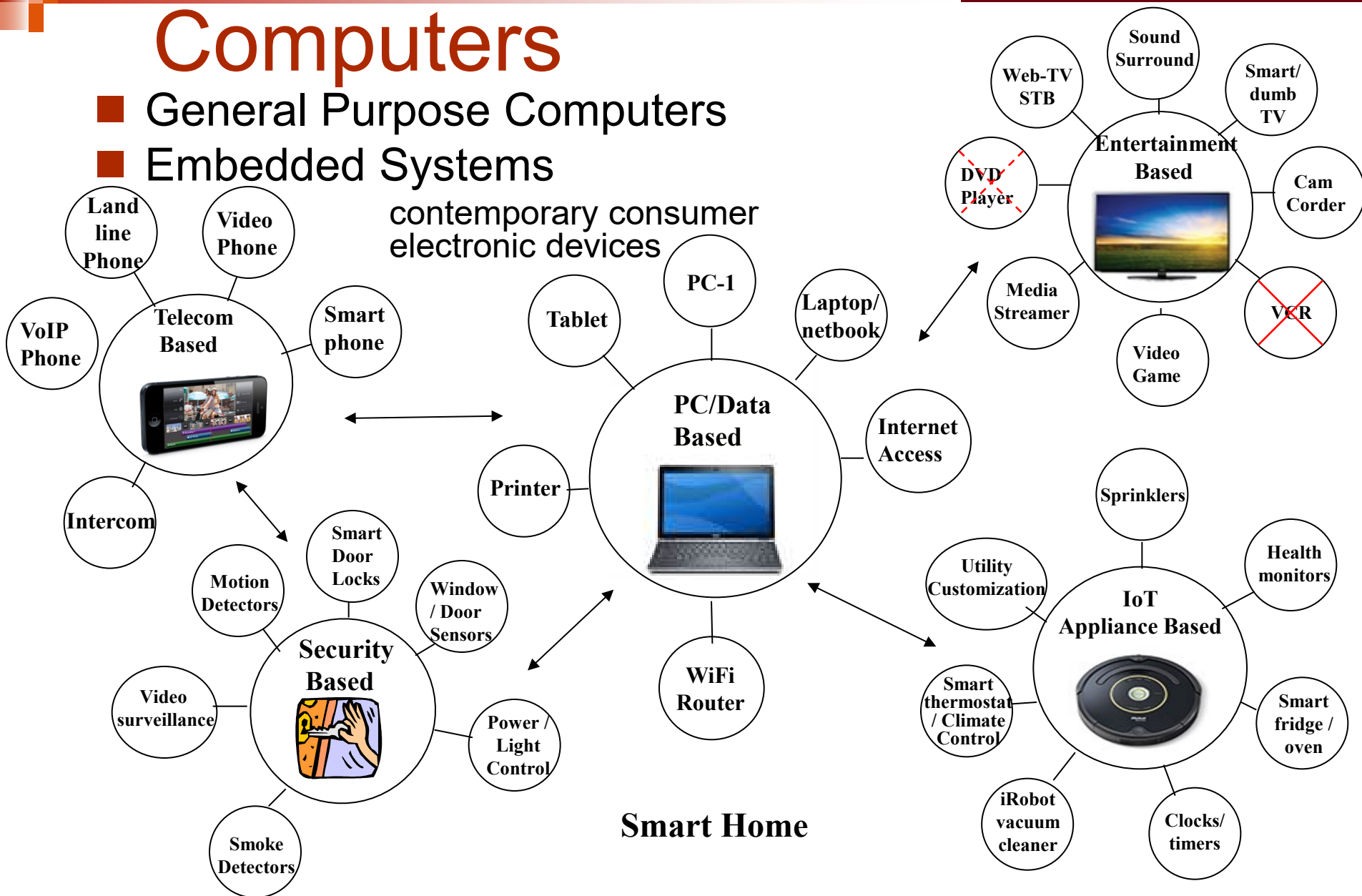
Microprocessor



Computers

- General Purpose Computers
- Embedded Systems

contemporary consumer
electronic devices



What do Computer Engineers do?

- Few Computer Engineers make a living designing computers from basic components.
- Even fewer Computer Engineers are involved in designing the integrated circuits that form the building blocks of computers.
- The majority of Computer Engineers configure and / or program existing **computer subsystems** in order to build more efficient products.
- Computer Engineers are expected to provide computer expertise (both software and hardware) to customers, to other engineers, to organizations and to their students.

Intro to CEG2x36 ...

- In this course HW&SW get together at the lowest level
- What will you learn in this course ?
 - How digital systems work
 - How digital systems are designed to execute SW programs
 - You will design many yourselves!
- Computer engineering is a very dynamic field
 - Chips double in speed every 18 months (Moore's Law), due to
 - Decreasing transistor sizes
 - Increasingly efficient computer architectures
- Performance / Power tradeoff
 - $\text{Performance} = (\text{frequency} * \text{IPS}) / \# \text{ of instructions}$
 - $\text{Power}_{\text{MOS}} = 0.5 * \text{capacitance} * \text{voltage}^2 * \text{frequency}$

Electron Tube
1950's



Discrete Transistor
1960's



Transistor dimension

Transistor Size

Integrated
Transistor

10 μm – 1971

6 μm – 1974

3 μm – 1977

1.5 μm – 1982

1 μm – 1985

800 nm – 1989

600 nm – 1994

350 nm – 1995

250 nm – 1997

180 nm – 1999

130 nm – 2001

90 nm – 2004

65 nm – 2006

45 nm – 2008

32 nm – 2010

22 nm – 2012

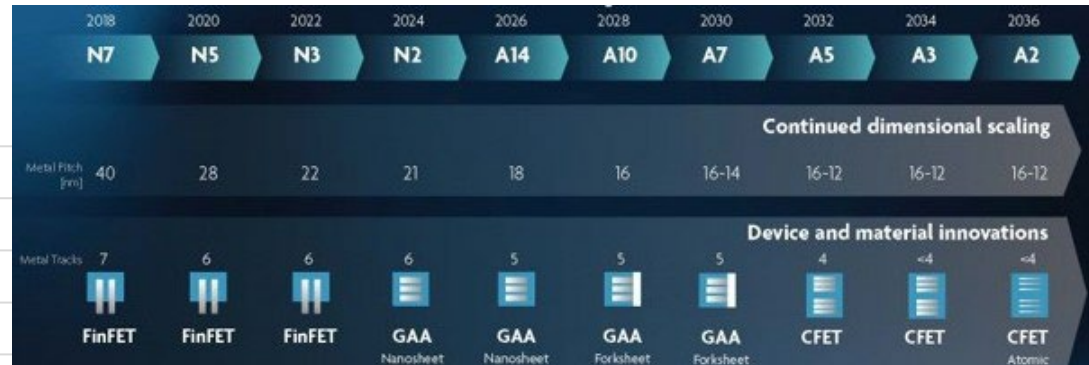
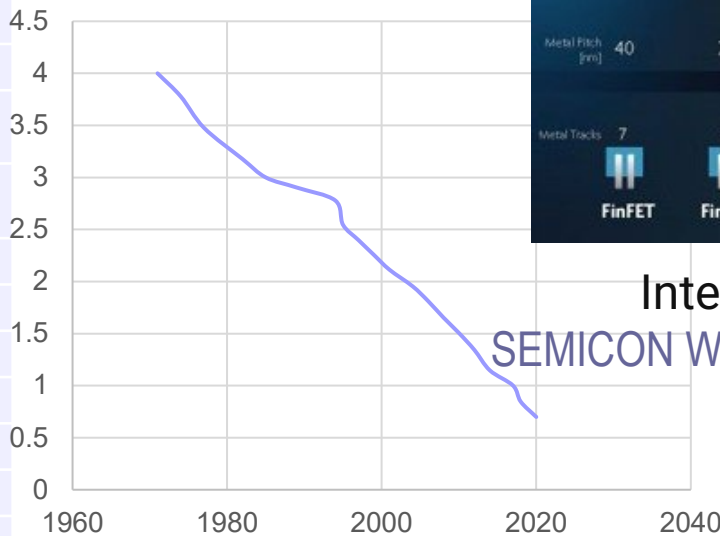
14 nm – 2014

10 nm – 2017

7 nm – 2018

5 nm – 2019

3 nm - 2021



Interuniversity Microelectronics Centre (IMEC):
SEMICON West 2022: [From Nanometers to the Angstrom Era](#)

GAA = Gate All Around

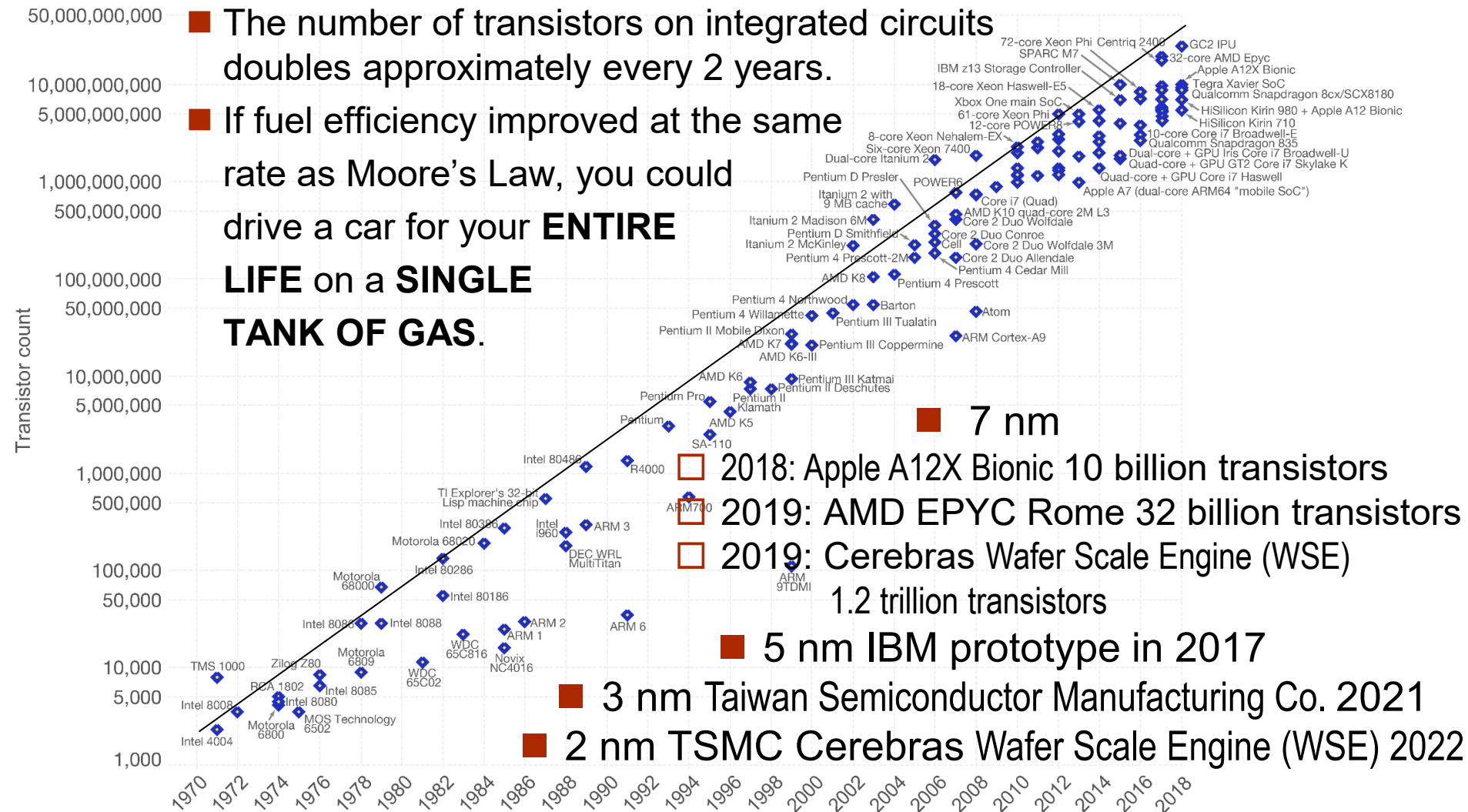
CFET = Complementary FET – vertical transistor

SIA (Semiconductor Industry Association) Roadmap

Years:	1999	2001	2003	2006	2008	2012	2014	2017	2018	2019	2021
Gate length (nm)	180	130	90	65	45	22	14	10	7	5	3
Transistors per cm^2 (millions)	14	16	24	40	64			100	126	173	300
Chip size (mm^2)	800	850	900	1000	1100	1300					

MOORE'S LAW

- The number of transistors on integrated circuits doubles approximately every 2 years.
- If fuel efficiency improved at the same rate as Moore's Law, you could drive a car for your **ENTIRE LIFE** on a **SINGLE TANK OF GAS**.



MOORE'S LAW: transistors / microprocessor

Date	Processor	Designer	Process	Area	Transistor count
1971	Intel 4004	Intel	10 μm	12 mm ²	2,300
1972	Intel 8008	Intel	10 μm	14 mm ²	3,500
1973	NEC $\mu\text{COM-4}$	NEC	7.5 μm	?	2,500
1973	Toshiba TLCS12	Toshiba	6 μm	32 mm ²	2,500
1974	Motorola 6800	Motorola	6 μm	16 mm ²	4,100
1974	Intel 8080	Intel	6 μm	20 mm ²	4,500
1976	Intel 8085	Intel	3 μm	20 mm ²	6,500
1976	Zilog Z80	Zilog	4 μm	18 mm ²	8,500
1978	Motorola 6809	Motorola	5 μm	21 mm ²	9,000
1978	Intel 8086	Intel	3 μm	33 mm ²	29,000
1979	Zilog Z8000	Zilog	μm		17,500
1979	Intel 8088	Intel	3 μm	33 mm ²	29,000
1979	Motorola 68000	Motorola	3.5 μm	44 mm ²	68,000
1981	WDC 65C02	WDC	3 μm	6 mm ²	11,500
1982	Intel 80186	Intel	3 μm	60 mm ²	55,000
1982	Intel 80286	Intel	1.5 μm	49 mm ²	134,000
1984	Motorola 68020	Motorola	2 μm	85 mm ²	190,000
1985	Intel 80386	Intel	1.5 μm	104 mm ²	275,000
1986	NEC V60 ^[26]	NEC	1.5 μm		375,000
1987	Motorola 68030	Motorola	0.8 μm	102 mm ²	273,000
1987	TI Lisp machine	TI	2 μm		553,000
1987	Hitachi Gmicro	Hitachi	1 μm		730,000
1988	Intel i960	Intel	1.5 μm		250,000
1989	Intel i860	Intel	μm		1,000,000
1989	Intel 80486	Intel	1 μm	173 mm ²	1,180,235
1990	68040	Motorola	0.65 μm	152 mm ²	1,200,000
1991	R4000	MIPS	1 μm	213 mm ²	1,350,000
1993	Hitachi HARP-1	Hitachi	0.5 μm	267 mm ²	2,800,000
1993	Pentium	Intel	0.8 μm	294 mm ²	3,100,000

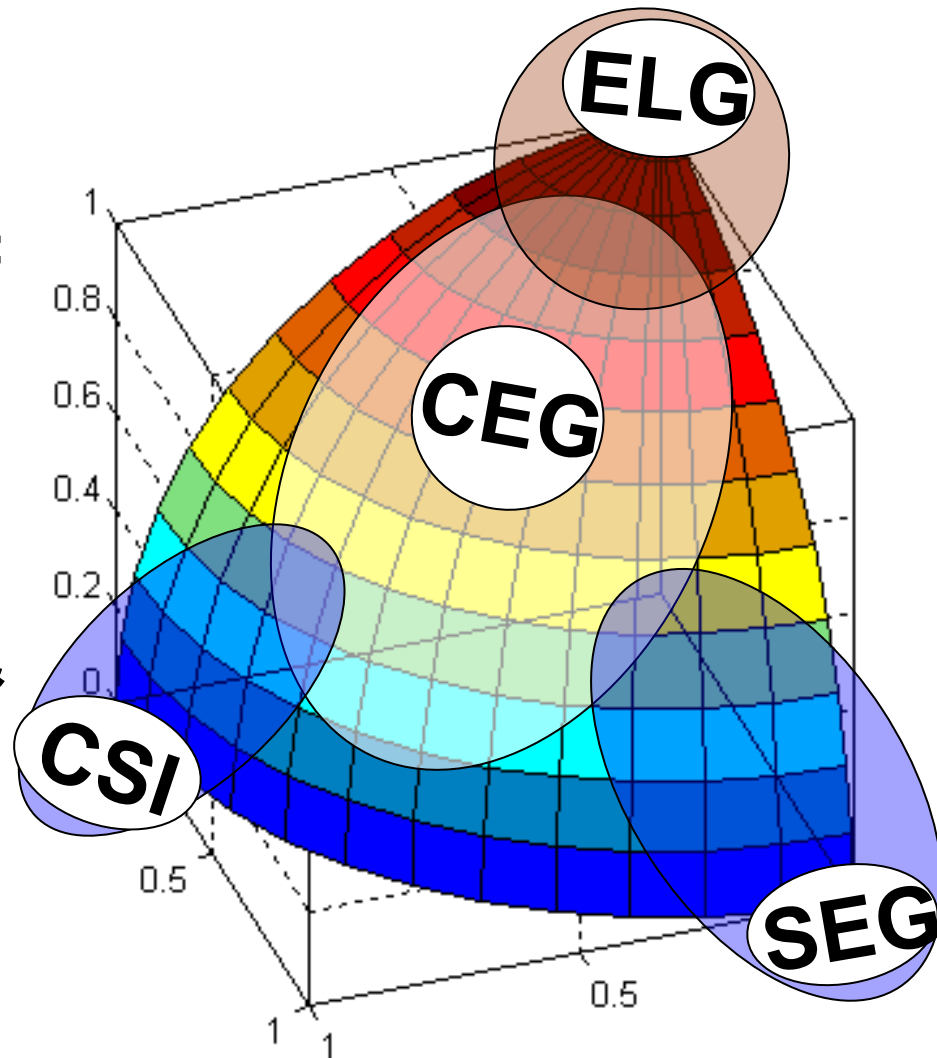
Date	Processor	Designer	Process	Area	Transistor count
1994	68060	Motorola	0.6 μm	218 mm ²	2,500,000
1995	Pentium Pro	Intel	0.5 μm	307 mm ²	5,500,000
1996	AMD K5	AMD	0.5 μm	251 mm ²	4,300,000
1997	Pentium II	Intel	0.35 μm	195 mm ²	7,500,000
1997	AMD K6	AMD	0.35 μm	162 mm ²	8,800,000
1997	Hitachi SH-4	Hitachi	0.2 μm	42 mm ²	10,000,000
1999	AMD K7	AMD	0.25 μm	184 mm ²	22,000,000
1999	Pentium II Mobile	Intel	0.18 μm	180 mm ²	27,400,000
2000	Pentium 4	Intel	0.18 μm	217 mm ²	42,000,000
2001	SPARC64 V	Fujitsu	0.13 μm	290 mm ²	191,000,000
2002	Itanium 2 McKinley	Intel	0.18 μm	421 mm ²	220,000,000
2003	Itanium 2 Madison	Intel	0.13 μm	374 mm ²	410,000,000
2004	Itanium 2	Intel	0.13 μm	432 mm ²	592,000,000
2006	Dual-core Itanium	Intel	90 nm	596 mm ²	1,700,000,000
2008	6 core Xeon 7400	Intel	45 nm	503 mm ²	1,900,000,000
2010	8-core Xeon	Intel	45 nm	684 mm ²	2,300,000,000
2011	10-core Xeon	Intel	32 nm	512 mm ²	2,600,000,000
2012	61-core Xeon Phi	Intel	22 nm	720 mm ²	5,000,000,000
2013	Xbox One SoC	AMD	28 nm	363 mm ²	5,000,000,000
2014	18-core Xeon	Intel	22 nm	661 mm ²	5,560,000,000
2015	32-core SPARC M7	Oracle	20 nm		10,000,000,000
2016	72-core Xeon Phi	Intel	14 nm	683 mm ²	8,000,000,000
2017	32-core AMD Epyc	AMD	14 nm	768 mm ²	19,200,000,000
2018	Apple A12X Bionic	Apple	7 nm	122 mm ²	10,000,000,000
2018	GC2 IPU	Graphcore	16 nm	825 mm ²	23,600,000,000
2019	AMD EPYC Rome	AMD	7 nm		32,000,000,000
2020	A100 AI	Nvidia	7 nm		54,000,000,000
2020	1872-core MK2 GC200 IPU	Graphcore	7 nm	825 mm ²	59,400,000,000
2020	Wafer Scale Engine (WSE)	Cerebras	7 nm		1,200,000,000,000

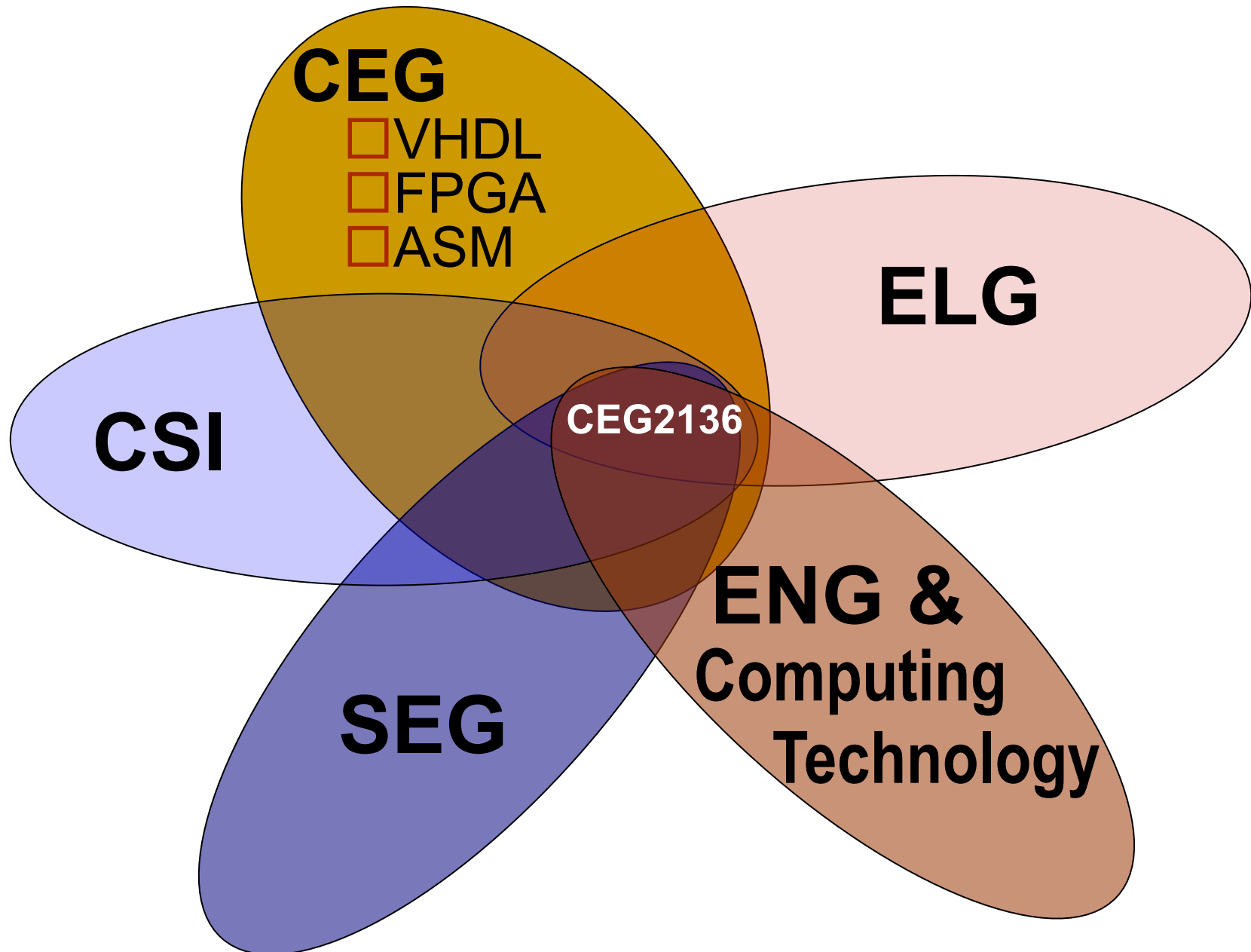
Study programs at School of EECS

Computer Engineering

requires mastery of both:

- *hardware* (overlapping with Electrical Eng.)
- *software* (overlapping with Computer Science & Software Engineering)





HARDWARE

ITI1100 Digital Systems I
Combinational & Sequential Logic Circuits

CEG2136 Computer Architecture I
ALU, CPU, Memory, computer organization

CEG3136 Computer Architecture II
Microprocessors, I/O, Embedded

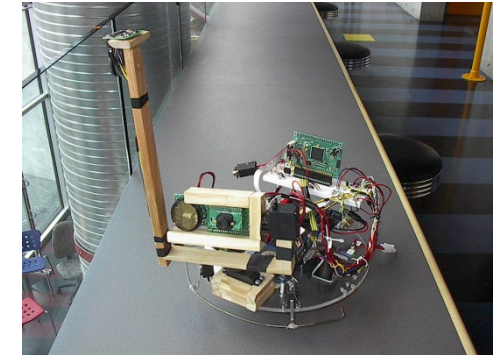
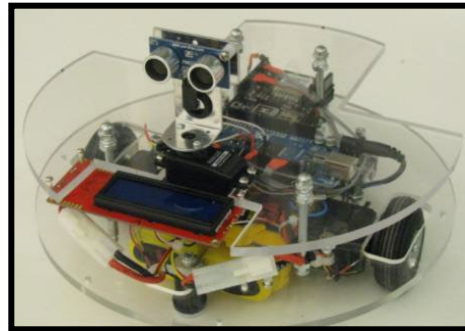
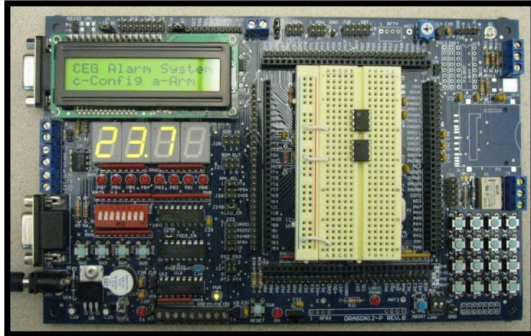
CEG3155 Digital Systems II
FSM, ASM, VHDL, PLD, FPGA

CEG3156 Computer Systems Design

CEG4136 Computer Architecture III
Multiprocessor Systems

**CEG4912 Computer Engineering
Design Project I**
**CEG4913 Computer Engineering
Design Project II**

Computer Engineering Design



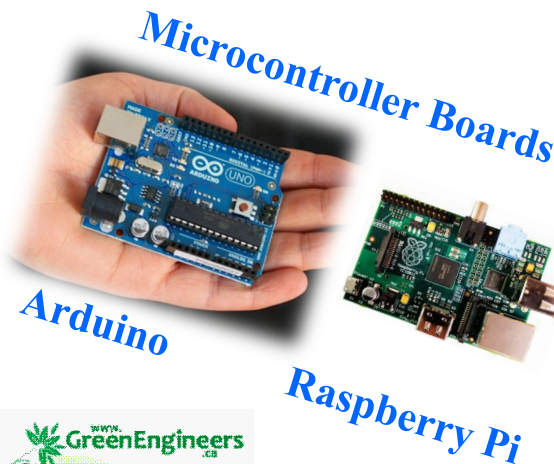
CEG3136 – Computer Architecture II

CEG4912/CEG4913 – Capstone Design Project

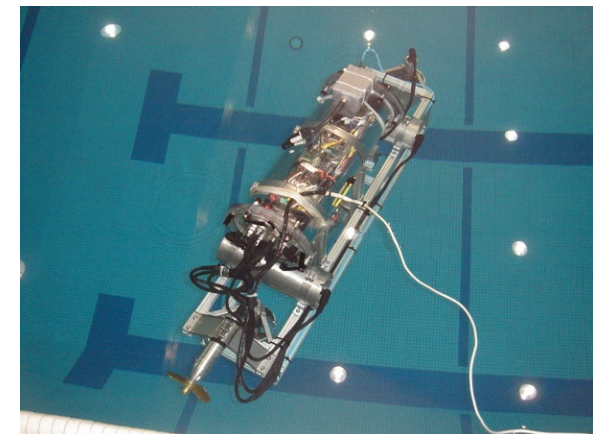
Unmanned Airplane (UAV)



Wind Turbine



Autonomous Underwater Vehicle

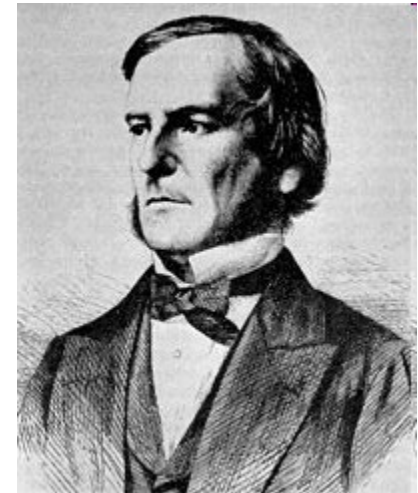


Logic Algebra

■ George Boole (1815 – 1864)

The genius inventor of the *Boolean Logic* was born in the wrong time!?

1847: “The Mathematical Analysis of Logic”



Claude Shannon (1916-2001)

Circuits with relays solve Boolean algebra problems

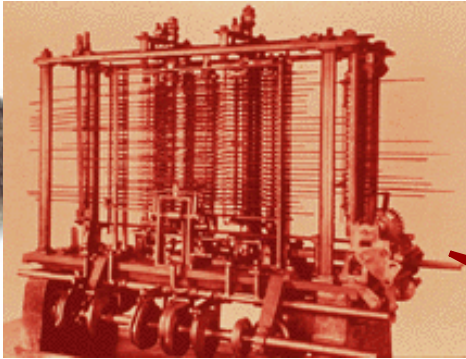
1937: EE master's thesis at the Massachusetts Institute of Technology

1938: “A Symbolic Analysis of Relay and Switching Circuits,” IEEE Transactions

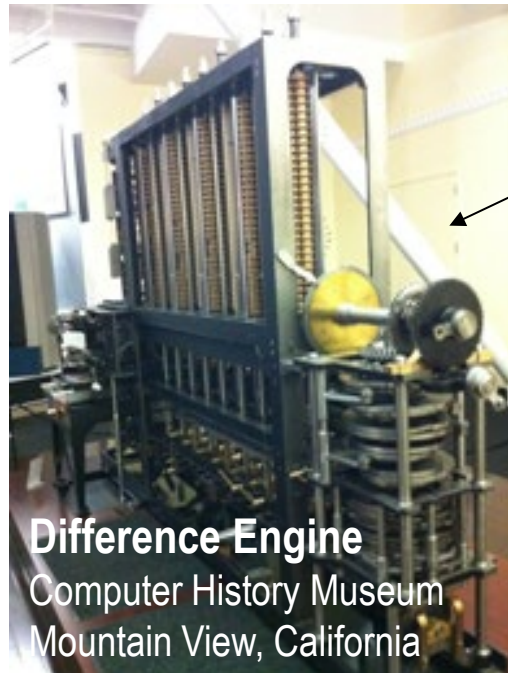




Charles Babbage (1791-1871)



- British mathematician: designs the first fully automatic calculating device, the Difference Engine (1822)
- 1837: First programmable calculator: steam-powered mechanical Analytical Engine, conceived but he never actually completed because of funding problems.

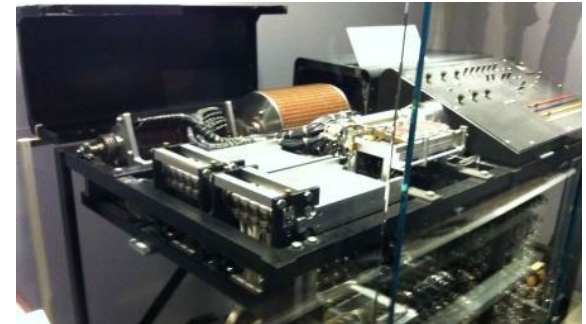
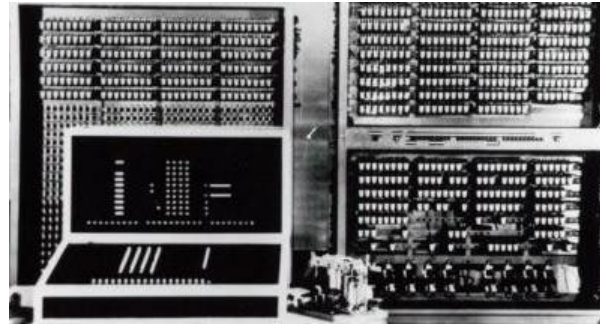


Difference Engine
Computer History Museum
Mountain View, California

The design included a CPU (“mill”, capable of conditional branching) and memory (“store”, holding up to 1000 50-digit numbers), and could read programs from punched cards.

- First programmers: Charles Babbage and Ada (daughter of Lord Byron).

First Computers...



Alan Turing, 1912-1954

- 1936: Turing Machine in "On Computable Numbers," Proc. of London Mathematical Society,
- 1941: the "Bombe"= electro-mechanical device to decipher German Enigma-machine - encrypted signals (WW II)

Konrad Zuse (1910-1995)

Z1 (1936)

- the first binary (electro-mechanical) programmable calculator / punched tape
- built in Zuse's parents' house

Z3 (1941)

- the first working **programmable**, binary, fully automatic **computer (relays)**.
- It was Turing-complete!
- statistical analyses of wing flutter.

John V. Atanasoff (1903-1995)

and his MSc student

Clifford Berry (1918-1963)

@ Iowa State University

Atanasoff-Berry Computer

(ABC) the first electronic digital computing device (but dedicated, **not programmable**) with DRAM!

- 1939: proof-of-concept prototype,
- 1942: the ABC is completed

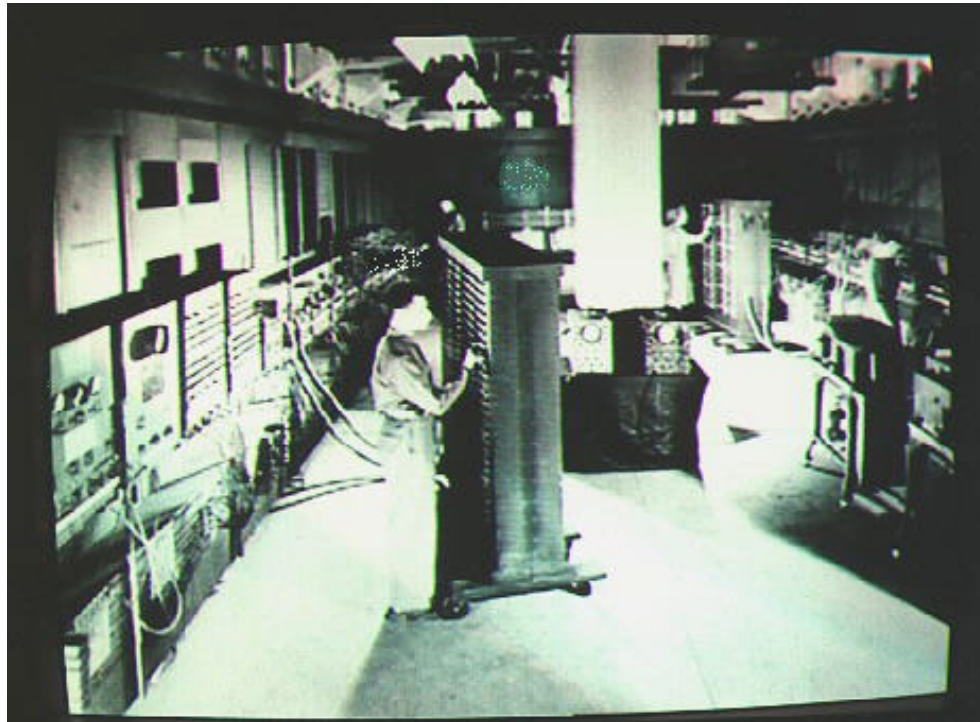
<http://jva.cs.iastate.edu/reconstructionoperation.php>

The **COMPUTER** as a concept was declared un-patentable and thus was freely open to all.

1973: Atanasoff vs Mauchly (i.e., Honeywell Company vs Sperry Rand Corporation)

ENIAC

Electrical Numerical Integrator And Calculator



John Mauchly & Presper Eckert
1946: University of Pennsylvania
the first **general purpose electronic computer** (base 10), to calculate
artillery firing tables for the US Army.

- Programmed by manipulating switches and cables
- Weighed 30 tons & used up an entire floor = 167 m²
- Dimmed the lights of the East side of Philadelphia
- 18,000 **vacuum tubes** - several tubes burned out almost every day

Von Neumann Architecture

■ 1945 - "First Draft of a Report on the **EDVAC**"

conceived by John Mauchly & Presper Eckert

- stored-program computer; base 2

■ 1946 - Summer school at the University of Pennsylvania

■ **Computer system** =

□ **CPU** - Central Processing Unit

■ **Arithmetic and Logic Unit**

(**ALU** + registers = DATA PATH)

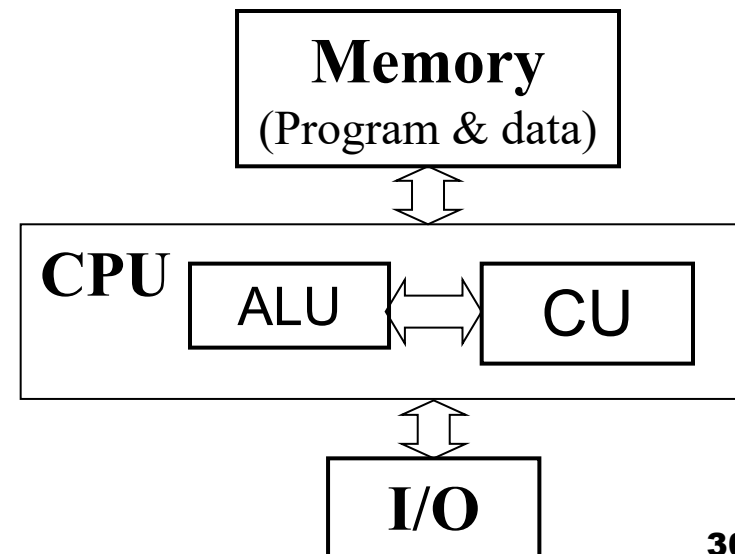
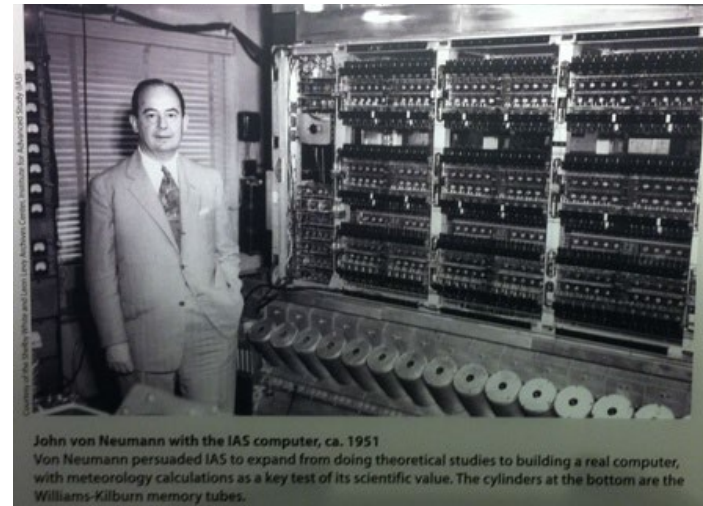
■ **CONTROL UNIT (CU)**

□ **Input / Output Unit (I/O)**

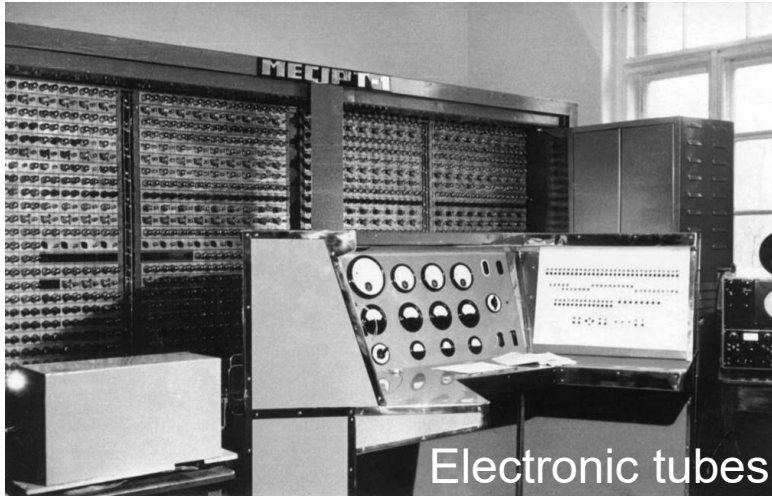
- Input Unit = Keyboards, mice
- Output Unit = printers, graphic displays

□ **Memory (M)**

- Primary (cache, RAM, HDD)
- Secondary (CD-ROM, SD)



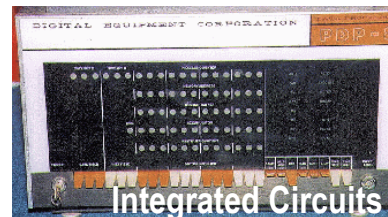
First Generation (1946-1959) Electronic Computer



Second Generation (1959-1965) Electronic Computer



Third Generation (1965-1971) IBM System/360



Third Generation 1965: DEC PDP8 the first minicomputer

Fourth Generation 1971: 1st microprocessor (Intel 4004 Federico Faggin) 1st DRAM (Intel 1103)

IBM Personal Computer

1981: XT / Intel 8088
1986: AT / Intel 80286



Fifth Generation : Artificial Intelligence (1982 & still under construction)

Home Computers

- family TV set
- BASIC interpreter + line editor
- K7 = external memory



1982: Commodore 64

- Jack Tramiel founded *CBM* in Toronto
- microprocessor 6510 /MOS Technology
- 17 million C64s over 1982 – 1994 - the best selling computer of all times

1982: Sinclair (UK) ZX Spectrum

- microprocessor Z80 / Zilog
- 5 million units worldwide (not counting clones, which were numerous)
- Over 20,000 SW titles have been released since its launch in 1982
- Most popular



1977:

Apple II by Steve Wozniak

- 1977 – 1993
- microprocessor 6510
MOS Technology



Hardware back in fashion!



33

- New York Times: “HW is becoming the new SW”
- **Raspberry Pi (UK)** - dimension of a credit card
 - single-board computer (SBC) to stimulating the teaching of basic computer science in schools
 - Modernized version of the ‘80’s ZX Spectrum or Commodor 64
 - Broadcom BCM2835 **SoC**, which includes
 - ARM processor ■ VideoCore GPU ■ 256 MB RAM
 - Linux / SD card ■ HDMI 1080p ■ USB ■ Ethernet
- **Arduino (Italy)** open source computer HW and SW company
- **Pebble: E-Paper Watch for iPhone & Android**
 - 2012: pledged of \$100,000 goal at **Kickstarter**
=> **68,929 backers -> \$10,266,846 in 1 month**
 - 2016: bought by Fitbit
 - 2021: Fitbit bought by Google
- **Smartphones**

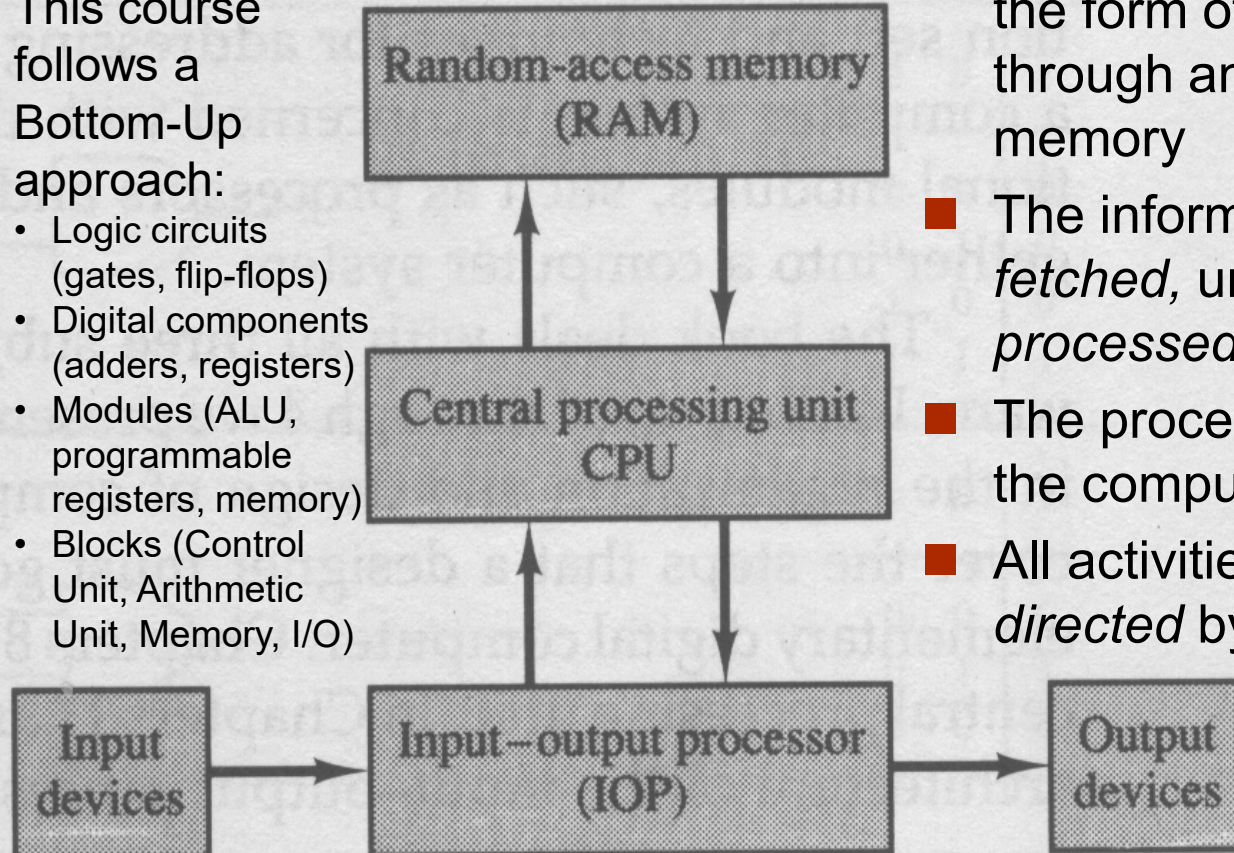


Basic Operation of a Computer

- Computer Architecture ($\approx$ digital computer organization) is concerned with the structure and operation of computers.
- Program: A program is a sequence of computer instructions.

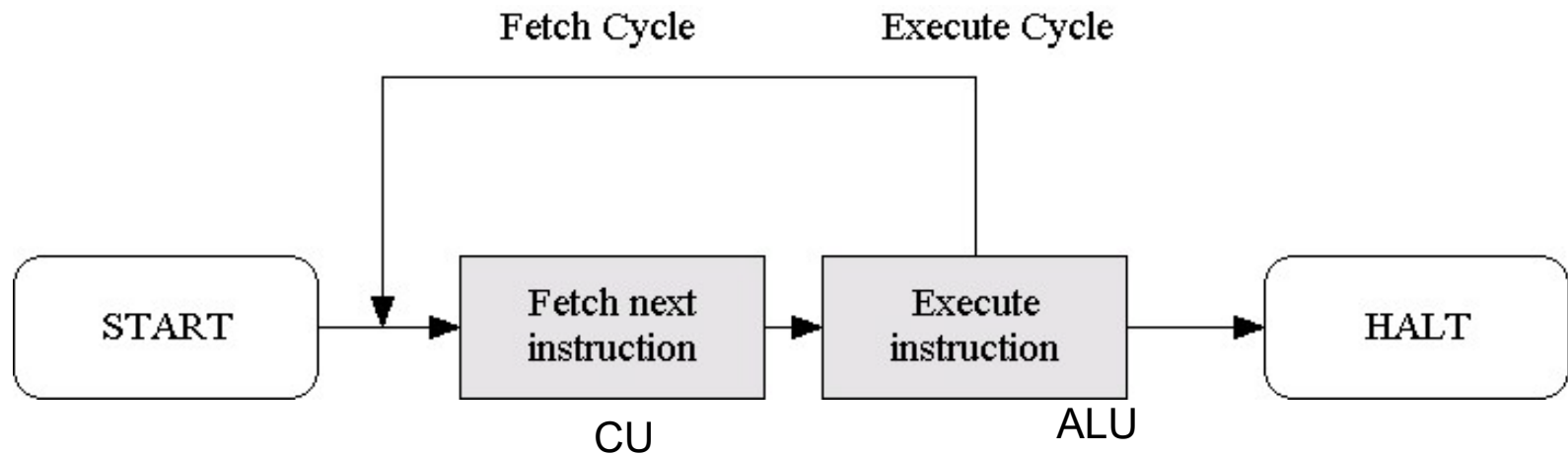
This course follows a Bottom-Up approach:

- Logic circuits (gates, flip-flops)
- Digital components (adders, registers)
- Modules (ALU, programmable registers, memory)
- Blocks (Control Unit, Arithmetic Unit, Memory, I/O)



- The computer *accepts* information in the form of programs and data through an input unit and *stores* it in memory
- The information stored in memory is *fetched*, under program control, and *processed* in an ALU
- The processed information *leaves* the computer through an output unit
- All activities inside the computer are *directed* by the control unit

Basic Instruction Cycle



- Computer performs the instruction cycle forever! (or at least until it is turned off, faces an error or is instructed to do stop)

